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Acquired *EGFR* C797S mediates resistance to AZD9291 in advanced non-small cell lung cancer harboring *EGFR* T790M

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Abstract

Here we studied cell-free plasma DNA (cfDNA) collected from subjects with advanced lung cancer whose tumors had developed resistance to the epidermal growth factor receptor (EGFR) tyrosine kinase inhibitor (TKI) AZD9291. We first performed next-generation sequencing of cfDNA from seven subjects and detected an acquired *EGFR* C797S mutation in one; expression of this mutant EGFR construct in a cell line rendered it resistant to AZD9291. We then performed droplet digital PCR on serial cfDNA specimens collected from 15 AZD9291-treated subjects. All were positive for T790M prior to treatment, but at resistance three molecular subtypes emerged: 6 cases acquired the C797S mutation, 5 cases maintained the T790M mutation but did not acquire the C797S mutation, and 4 cases lost the T790M mutation despite detecting of the underlying *EGFR* activating mutation. Our findings provide insight into the diversity of mechanisms through

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Author contributions

K.S.T, P.A.J, and G.R.O. designed the study. E.F., B.C.C., S.M., M.C., P.A.J., and G.R.O. enrolled patients and provided specimens. C.P.P., D.S., B.D., Z.L., A.M., A.V., Y.K., D.E., and J.C.B. performed and/or designed experiments. K.S.T. and G.R.O. drafted the manuscript. All authors reviewed the final manuscript.

Author disclosures

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which tumors acquire resistance to AZD9291 and highlight the need for therapies able to overcome resistance mediated by *EGFR* C797S.

EGFR-mutant lung cancer is a subtype of non-small cell lung cancer (NSCLC) that exhibits sensitivity to *EGFR* TKIs such as erlotinib and gefitinib; however, acquired resistance develops after a median of 9–14 months^{1,2}. The most common mechanism of TKI resistance is a second-site mutation (T790M) in the *EGFR* kinase domain, which can be detected in >50% of biopsies done after resistance develops^{3,4}. AZD9291 is an oral, irreversible, mutant-selective *EGFR* TKI developed to have potency against tumors bearing *EGFR* activating mutations (e.g. L858R or exon 19 deletion) in the presence of T790M^{5–7}. In the ongoing phase I AURA study, AZD9291 induced durable responses in *EGFR*-mutant lung cancer patients with acquired resistance to other *EGFR* TKIs, with preliminary progression free survival estimates of ~10 months in T790M+ patients⁸.

To identify potential mechanisms of resistance to AZD9291 prior to the availability of resistance biopsy specimens, we studied cfDNA which was collected during the phase I study AURA. We first performed next-generation sequencing (NGS) on cfDNA from 7 subjects whose disease progressed and from whom paired pre-treatment and post-disease progression plasma specimens were available. cfDNA was isolated from plasma and all exons of a 20 gene panel were PCR amplified and analyzed using an Illumina HiSeq.

One subject (Subject #1) was a 33 year-old female whose cancer had progressed on multiple prior lines of chemotherapies and *EGFR* TKIs; rebiopsy was T790M+. After 6 weeks of AZD9291 treatment, scans demonstrated a partial response (Supplementary Fig. 1); however, she developed systemic progression after 23 weeks of AZD9291. NGS of plasma collected from this subject at time of systemic progression revealed a new C797S mutation in exon 20 of *EGFR* in addition to the exon 19 deletion and T790M mutations present before treatment with AZD9291 (Fig. 1a).

Based on *in vitro* studies, the *EGFR* C797S mutation is thought to induce resistance to irreversible *EGFR* TKIs, including quinazoline-based compounds (e.g. HKI-272) and pyrimidine-based compounds (e.g. WZ4002), by impairing covalent binding of these drugs to the *EGFR* protein^{5,9–11}. To confirm that C797S induces resistance to AZD9291, we generated Ba/F3 cells stably expressing an *EGFR* activating mutation (exon 19 deletion or L858R) and T790M in *cis* either with or without the C797S mutation. Cells expressing the C797S-mutant construct were markedly less sensitive to AZD9291 in terms of cell growth and *EGFR* phosphorylation (Figs. 1b–c, Supplementary Fig. 2); they were similarly resistant to CO-1686, a second mutant-selective *EGFR* TKI which has induced responses in T790M+ lung cancer¹². We therefore hypothesized that *EGFR* C797S could be a common mediator of acquired resistance to AZD9291 in patients.

To confirm the plasma NGS findings, we developed a droplet digital PCR (ddPCR) assay as done previously for detection of other *EGFR* mutations in cfDNA¹³. ddPCR of serial plasma specimens from Subject #1 confirmed a high plasma concentration of exon 19 deletion and T790M prior to treatment, without evidence of C797S (Fig. 1d). The concentration of the exon 19 deletion and T790M mutations reduced 100-fold at week 6, and increased as the

cancer developed systemic progression (weeks 12–23); at progression, a newly acquired *EGFR* C797S mutation was detected. We then performed serial ddPCR profiling on a total of 19 subjects with advanced *EGFR*-mutant NSCLC with acquired resistance to AZD9291 (two of these 19 subjects were also in the original NGS cohort). Pretreatment plasma ddPCR detected *EGFR* activating mutations in all subjects, T790M in 15 subjects, and C797S in no subjects (Supplementary Table 1). In the 15 T790M+ cases, plasma C797S was detected at progression in 6 (40%), always with a detectable plasma T790M (Fig. 1d), and all harboring an exon 19 deletion as their *EGFR* activating mutation. In another 5 of the 15 T790M+ cases (33%), the T790M mutation was again detected at progression without evidence of C797S (Fig. 1e). Intriguingly, in 4 of the 15 T790M+ cases (27%), the T790M mutation was no longer detectable at progression despite high levels detected prior to AZD9291 treatment (Fig. 1f); in such cases the *EGFR* activating mutation nevertheless increased in abundance at time of progression. Finally, in the four cases that were T790M– prior to AZD9291 treatment, none had detectable C797S or T790M in plasma at progression (Supplementary Table 1).

We performed plasma NGS on cases with acquired *EGFR* C797S to further characterize this mutation. In Subject #1, individual sequencing reads contained both the T790M and C797S mutations, indicating that they occur on the same allele (Fig. 1a). In another case (Subject #2), NGS of progression plasma detected T790M and C797S mutations on different alleles (Fig. 2a), suggesting that T790M– alleles also can acquire C797S. In two cases, tumor biopsies following progression on AZD9291 were available (Subjects #4 and #5), and targeted NGS of these confirmed an acquired C797S mutation that was not detected in the pretreatment tumors (Figs. 2b–c), and no other acquired mutations. NGS of plasma obtained after disease progression in these subjects identified the same DNA alteration seen in the tumor and also identified a second DNA alteration encoding for C797S (Figs. 2b–c), suggesting that multiple clones in parallel may be able to acquire this resistance mutation. No other acquired mutations were identified in any of the other cases or genes studied using plasma NGS.

This genomic analysis of serial cfDNA specimens from an ongoing first-in-man study of AZD9291 identified three molecular subtypes of acquired resistance including one defined by an *EGFR* C797S mutation not previously detected in patient samples. This acquired *EGFR* C797S mutation is anticipated to induce resistance to all covalent EGFR TKIs, highlighting the need for novel strategies to inhibit EGFR even in the presence of this mutation. EGFR C797S mechanistically parallels the acquired Bruton tyrosine kinase (BTK) C481S mutation seen in lymphoma following progression on irreversible BTK inhibitors¹⁴, suggesting that these cysteine point mutations that block drug binding may be a recurring vulnerability for a broad range of covalent kinase inhibitors. While a more comprehensive analysis of plasma and biopsy tissue collected following resistance will be needed to provide greater clarity on C797S incidence, the emergence of C797S in a marked proportion of AZD9291-resistant patients suggests that development of targeted therapies with the ability to overcome C797S is warranted. C797S was only identified in cases harboring an exon 19 deletion (6 of 9) and not in cases harboring L858R (0 of 6); because our cell line data suggests C797S induces resistance regardless of sensitizing mutation (Fig. 1b), we suspect

this finding is due to the small sample size studied, though a differential disposition towards developing C797S will need to be considered going forward.

Our finding that some cancers treated with AZD9291 convert from T790M+ to T790M-, and our identification of cases harboring two concurrent acquired C797S mutations, suggests an underappreciated genomic heterogeneity associated with resistance to EGFR TKIs in NSCLC. Such heterogeneity requires further study, but may indicate a need for combination therapies which can inhibit or prevent the emergence of multiple resistance mechanisms simultaneously. The emerging toxicity profile of AZD9291 makes it potentially suitable for such rational combinations, some of which are currently under investigation in a multi-arm phase I study (NCT02143466) combining AZD9291 with either an anti-PD-L1 antibody, a MET inhibitor, or a MEK inhibitor.

Online Methods

Trial design

The AURA study (NCT01802632) is a phase I study of AZD9291 which enrolled subjects with *EGFR*-mutant lung cancer and acquired resistance to *EGFR* TKI to escalating doses of AZD9291 (20–240mg)⁸. At each dose level, expansion cohorts specifically studied patients with *EGFR* T790M+ or T790M- tumors based on a central genotyping assay. The study received IRB approval at all participating centers, and all patients provided informed consent prior to treatment. On this study, objective tumor response was assessed using the Response Evaluation Criteria In Solid Tumors (RECIST)¹⁵.

For the initial exploratory NGS cohort, we selected 7 subjects from the AURA trial (4 male, 3 female, median age 52, range 33–65) based upon availability of pretreatment and post-progression plasma specimens, regardless of evidence of sensitivity to therapy. We then performed the confirmatory plasma analysis on subjects meeting criteria for acquired resistance to AZD9291, defined as one scan demonstrating response or stable disease on imaging, followed by development of systemic progression on treatment. From this population, we only studied those with paired pretreatment and end-of-treatment plasma specimens available as of 9/22/2014. We further limited the analysis to subjects harboring *EGFR* L858R or exon 19 deletions, excluding subjects with rare activating mutations not detectable with existing ddPCR assays. We also omitted from the resistance analysis those subjects with no detectable L858R or exon 19 deletion in cfDNA at time of progression due to the probability of inadequate cfDNA shed to allow genomic studies to be performed. Nineteen subjects met these eligibility criteria and were included in the confirmatory plasma analysis (5 male, 14 female, median age 56, range 28–72).

Collection of plasma and extraction of cell-free DNA (cfDNA)

Per the AURA study protocol, plasma was collected for all subjects on study when feasible to allow for genomic analysis of cfDNA. 10–20 cc of blood were collected into EDTA tubes pretreatment, intermittently on treatment (dependent on the specific cohort), and again when coming off AZD9291. In a parallel study at the Dana-Farber Cancer Institute (DFCI), we additionally consented subjects on the same clinical trial to collection of specimens every 3 weeks for more detailed analysis of response and progression kinetics.

Within 4 hours of collection, whole blood was centrifuged for 10 min at 1200g after which the plasma supernatant was further cleared by centrifugation for 10 min at 3000g. Cleared plasma was stored in cryostat tubes at -80°C until use. Cell free DNA was isolated using the QIAmp Circulating Nucleic Acid Kit (Qiagen) according to the manufacturer's protocol. DNA was eluted in AVE buffer (100 uL) and stored at -80 °C.

NGS of plasma cfDNA

For plasma NGS, we PCR amplified all coding exons of a 20 gene panel using the Qiagen GeneRead Lung Cancer version 1 kit (catalog # 180941 NGHS-005Z-96); genes in this panel are listed in Supplementary Table 2. We prepared libraries of amplified DNA with the Kapa Hyperprep kit (catalog # KK8502). We then performed ultra-deep sequencing on an Illumina HiSeq instrument (2×100 bp) following QC with an Agilent TapeStation. We analyzed NGS data within the BCBio framework (<https://bcbio-nextgen.readthedocs.org/en/latest/>) using an AstraZeneca-developed variant calling algorithm (<https://github.com/AstraZeneca-NGS/VarDict>). NGS was performed once on each specimen. Given the difficulty of accurate variant calling when using plasma-derived cfDNA and ultra-deep coverage of genes of interest (average 30,000×), we manually inspected variant calls of interest in IGV. We compared *EGFR* activating and resistance mutations to orthogonal genotyping data from plasma and tissue.

C797S cell line studies

We introduced mutant *EGFR* constructs into authenticated Ba/F3 cells by retroviral infection as previously described^{5,16}. Parental Ba/F3 cell lines were obtained previously from the laboratory of James Griffin at DFCI. We assessed growth and inhibition of growth by MTS assay performed according to previously established methods.^{5,16} We exposed mutant Ba/F3 cells to drug treatment for 72 hours and plotted the data relative to non-drug treated cells with the same genotype. We set up all experimental points in six wells and repeated all experiments three times. We graphically displayed the data using GraphPad Prism version 5.0 for Windows, (GraphPad Software; www.graphpad.com), fitting the curves using a non-linear regression model with a sigmoidal dose response.

Following drug treatment for 6 hours, we lysed Ba/F3 cells in NP-40 buffer (Cell Signaling Technology). We conducted Western blot analyses after separation by SDS/PAGE electrophoresis and transfer to polyvinylidene difluoride-P membrane (Millipore). We performed immunoblotting according to the antibody manufacturers' recommendations and detected antibody binding using an enhanced chemiluminescence system (Perkin Elmer Inc.). The phospho-specific *EGFR* (pY1068) antibody was purchased from Cell Signaling (#3777s). Total *EGFR* and Tubulin antibodies were purchased from Bethyl Laboratories (#A300-388A) and Sigma (#T9026), respectively.

Droplet digital PCR of plasma cfDNA

We developed *EGFR* C797S ddPCR assays in a similar fashion as previously described for other *EGFR* mutations¹³. In brief, we custom designed primer and probe pairs, optimizing them for annealing temperature and cycling condition using serial dilutions of mutant DNA we obtained from Ba/F3 cells with and without *EGFR* T790M/C797S. Similar to our

previously developed ddPCR assays, the *EGFR* C797S ddPCR assays achieved a sensitivity between 0.05% and 0.1% (Supplementary Figs. 3–4). We purchased all ddPCR reagents from Bio-Rad, and custom ordered primers and probes from Life Technologies. We performed the ddPCR analysis on a BioRad QX100 ddPCR instrument as previously described¹³, with following PCR conditions:

EGFR L858R: forward primer, 5'-GCAGCATGTCAAGATCACAGATT-3', reverse primer, 5'-CCTCCTTCTGCATGGTATTCTTTCT-3'. Probe sequences: 5'-VIC-AGTTTGGCCAGCCCAA-MGB-NFQ-3', 5'-FAM-AGTTTGGCCCGCCCAA-MGB-NFQ-3'. Cycling conditions: 95 °C × 10 min (1 cycle), 40 cycles of 94 °C × 30 s and 58 °C × 1 min, and 10 °C hold.

EGFR del 19: forward primer, 5'-GTGAGAAAGTTAAAATTCCCGTC-3', reverse primer, 5'-CACACAGCAAAGCAGAAAC-3'. Probe sequences: 5'-VIC-ATCGAGGATTTCCTTGTTG-MGB-NFQ-3', 5'-FAM-AGGAATTAAGAGAAGCAACATC-MGB-NFQ-3'. Cycling conditions: 95 °C × 10 min (1 cycle), 40 cycles of 94 °C × 30 s and 55 °C × 1 min, followed by 10 °C hold.

EGFR T790M: forward primer, 5'-GCCTGCTGGGCATCTG-3', reverse primer, 5'-TCTTTGTGTTCCCGGACATAGTC-3'. Probe sequences are: 5'-VIC-ATGAGCTGCGTGATGAG-MGB-NFQ-3', 5'-FAM-ATGAGCTGCATGATGAG-MGB-NFQ-3'. Cycling conditions: 95 °C × 10 min (1 cycle), 40 cycles of 94 °C × 30 s and 58 °C × 1 min, followed by 10 °C hold.

EGFR C797S (T>A): forward primer: 5'-GCCTGCTGGGCATCTG-3', reverse, 5'-TCTTTGTGTTCCCGGACATAGTC-3'. Probe sequences are: 5'-VIC-TTCGGCTGCCTCCTG-MGB-NFQ-3', 5'-FAM-TTCGGCAGCCTCC-MGB-NFQ-3'. Cycling conditions: 95 °C × 10 min (1 cycle), 40 cycles of 94 °C × 30 s and 56 °C × 1 min, followed by 10 °C hold.

EGFR C797S (G>C): forward primer: 5'-GCCTGCTGGGCATCTG-3', reverse, 5'-TCTTTGTGTTCCCGGACATAGTC-3'. Probe sequences are: 5'-VIC-TTCGGCTGCCTCCTG-MGB-NFQ-3', 5'-FAM-CTTCGGCTCCCTCCTG-MGB-NFQ-3'. Cycling conditions: 95 °C × 10 min (1 cycle), 40 cycles of 94 °C × 30 s and 56 °C × 1 min, followed by 10 °C hold.

We ran each ddPCR assay in triplicate, each using 1/20th of the DNA isolated from 2mL plasma. We analyzed the results using QuantaSoft (ver 1.6.6) software that accompanied the QX100 reader. In our analyses, *EGFR* mutation specific signals are generated in the FAM channel while the *EGFR* wildtype signals are in the VIC channel. We dynamically determined cut-off values for positive signals using cell lysate controls with the respective *EGFR* mutations run concomitantly. We calculated poisson concentrations using QuantaSoft, and *EGFR* mutation concentrations were adjusted to mL of plasma as follows: * 3.33. %mut were calculated from generated Poisson concentrations as follows:

$$\%mut = \frac{[FAM]}{[FAM + VIC]} * 100$$
 Serial ddPCR measurements were plotted from start of treatment with AZD9291 until time of systemic progression.

Next-generation sequencing of resistance tumor biopsies

We extracted DNA from 5×10 um sliced sections of FFPE material using the Maxwell FFPE Tissue LEV DNA Purification Kit. Tumor area content was evaluated by a clinical pathologist to confirm adequacy for sequencing; a minimum tumor content was set to 20%, in order to better detect somatic mutations. An initial multiplex-PCR with a proof-reading polymerase was performed on samples. For NGS, Vall d'Hebron Institute of Oncology has developed a panel of over 600 primer pairs targeting frequent mutations in oncogenes plus several tumor suppressors, totaling 57 genes, including *EGFR*; see Supplementary Table 2 for a list of genes included. We pooled indexed libraries and sequenced then on a MiSeq instrument (2×100) at an average coverage of 3000×. Initial alignment was performed with BWA after primer sequence clipping and variant calling was done with the GATK Unified Genotyper and VarScan2 followed by ANNOVAR annotation. Mutations were called at a minimum 3% allele frequency. SNPs were filtered out with dbSNP and 1000 genome datasets. All detected variants were manually checked.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

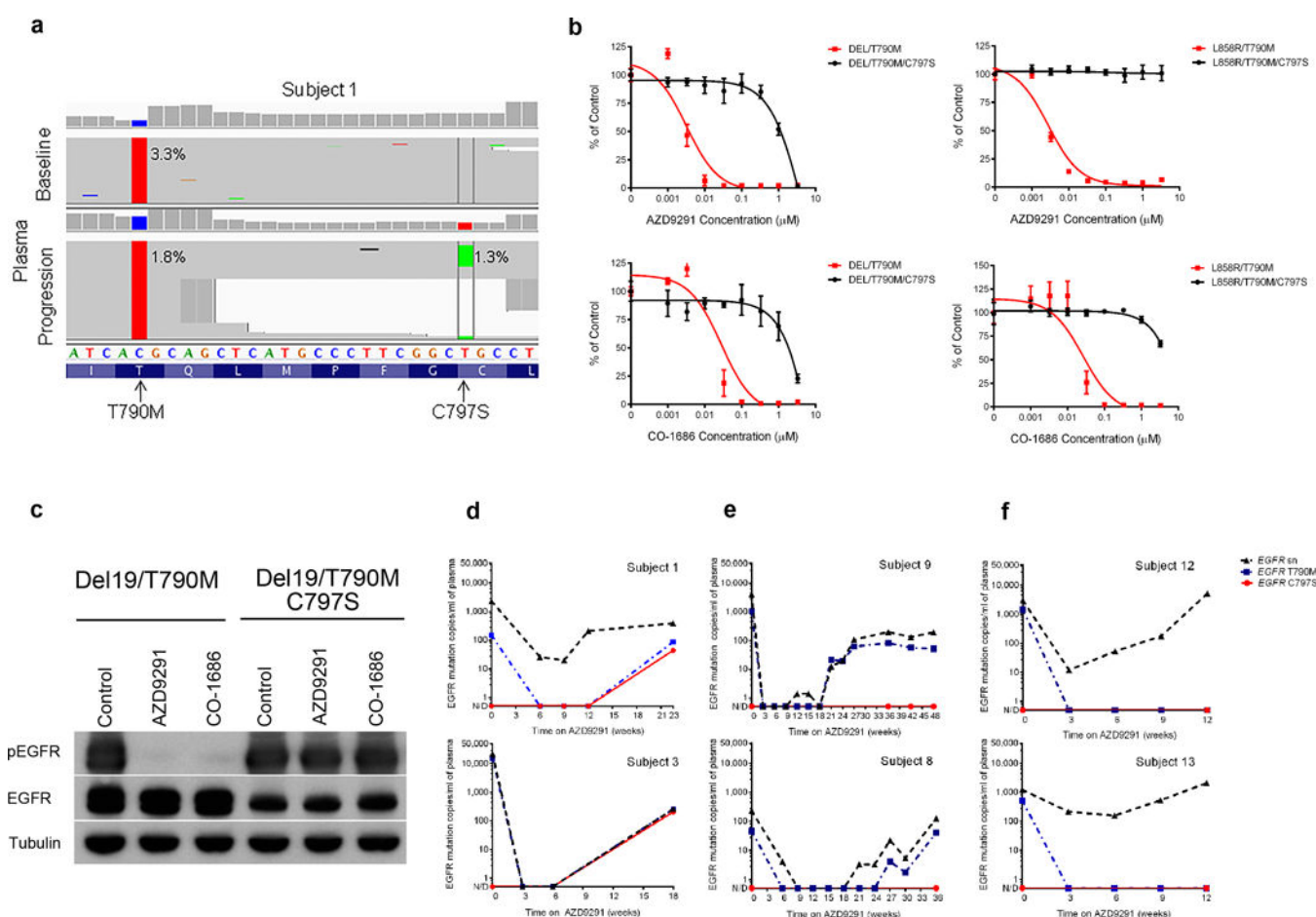
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References

1. Mok TS, et al. Gefitinib or carboplatin-paclitaxel in pulmonary adenocarcinoma. *N Engl J Med*. 2009; 361:947–957. [PubMed: 19692680]
2. Rosell R, et al. Screening for epidermal growth factor receptor mutations in lung cancer. *N Engl J Med*. 2009; 361:958–967. [PubMed: 19692684]
3. Arcila ME, et al. Rebiopsy of Lung Cancer Patients with Acquired Resistance to EGFR Inhibitors and Enhanced Detection of the T790M Mutation Using a Locked Nucleic Acid-Based Assay. *Clin Cancer Res*. 2011; 17:1169–1180. [PubMed: 21248300]
4. Sequist LV, et al. Genotypic and Histological Evolution of Lung Cancers Acquiring Resistance to EGFR Inhibitors. *Sci Transl Med*. 2011; 3:75ra26.
5. Zhou W, et al. Novel mutant-selective EGFR kinase inhibitors against EGFR T790M. *Nature*. 2009; 462:1070–1074. [PubMed: 20033049]
6. Ward RA, et al. Structure- and Reactivity-Based Development of Covalent Inhibitors of the Activating and Gatekeeper Mutant Forms of the Epidermal Growth Factor Receptor (EGFR). *J Med Chem*. 2013; 56:7025–7048. [PubMed: 23930994]
7. Cross DAE, et al. AZD9291, an irreversible EGFR TKI, overcomes T790M-mediated resistance to EGFR inhibitors in lung cancer. *Cancer Discovery*. 2014
8. Jänne PA, et al. AZD9291 in EGFR Inhibitor Resistant Non-Small Cell Lung Cancer. *N Engl J Med*. 2015
9. Yu Z, et al. Resistance to an Irreversible Epidermal Growth Factor Receptor (EGFR) Inhibitor in EGFR-Mutant Lung Cancer Reveals Novel Treatment Strategies. *Cancer Res*. 2007; 67:10417–10427. [PubMed: 17974985]

10. Godin-Heymann N, et al. The T790M “gatekeeper” mutation in EGFR mediates resistance to low concentrations of an irreversible EGFR inhibitor. *Mol Cancer Ther.* 2008; 7:874–879. [PubMed: 18413800]
11. Avizienyte E, Ward RA, Garner AP. Comparison of the EGFR resistance mutation profiles generated by EGFR-targeted tyrosine kinase inhibitors and the impact of drug combinations. *Biochem J.* 2008; 415:197–206. [PubMed: 18588508]
12. Sequist LV, et al. Rociletinib in EGFR mutant Non Small-Cell Lung Cancer. *N Engl J Med.* 2015
13. Oxnard GR, et al. Noninvasive detection of response and resistance in EGFR-mutant lung cancer using quantitative next-generation genotyping of cell-free plasma DNA. *Clin Cancer Res.* 2014; 20:1698–1705. [PubMed: 24429876]
14. Woyach JA, et al. Resistance mechanisms for the Bruton’s tyrosine kinase inhibitor ibrutinib. *N Engl J Med.* 2014; 370:2286–2294. [PubMed: 24869598]
15. Eisenhauer EA, et al. New response evaluation criteria in solid tumours: revised RECIST guideline (version 1.1). *Eur J Cancer.* 2009; 45:228–247. [PubMed: 19097774]
16. Engelman JA, et al. PF00299804, an Irreversible Pan-ERBB Inhibitor, Is Effective in Lung Cancer Models with EGFR and ERBB2 Mutations that Are Resistant to Gefitinib. *Cancer Res.* 2007; 67:11924–11932. [PubMed: 18089823]

**Fig. 1.**

Acquired resistance to AZD9291 mediated by acquired *EGFR* C797S. **(a)** In the index case (Subject #1), targeted NGS identified an acquired T→A mutation (green) in 1.3% of reads, encoding for an *EGFR* C797S mutation. Overlapping reads spanning T790 and C797 contain both the T790M and C797S mutations, indicating the two mutations occur in *cis* on the same allele. **(b)** Ba/F3 cells harboring one of two *EGFR* activating mutations (exon 19 deletion or L858R) plus the T790M resistance mutation, either with or without C797S, were treated with either AZD9291 or CO-1686 at the indicated concentrations, and viable cells were measured after 72 hours of treatment and plotted relative to untreated control cells. Experiments were repeated 3 times, with mean and standard deviation plotted at each concentration. The curves were fitted using a non-linear regression model with a sigmoidal dose response. **(c)** Ba/F3 cells expressing *EGFR* del 19/T790M and del 19/T790M/C797S cells were treated with 1.0 μM AZD9291 or CO-1686 for 6 hours. Cell extracts were immunoblotted to detect total or phosphorylated *EGFR* and tubulin (loading control). **(d–f)** Representative images from serial plasma ddPCR show three molecular subtypes of acquired resistance to AZD9291 (N/D: not detected). A subset of subjects acquire an *EGFR* C797S resistance mutation, always in the presence of T790M **(d)**. Other subjects maintain the *EGFR* T790M mutation without evidence of an acquired C797S **(e)**. The remaining

subjects lose the *EGFR* T790M mutation despite increasing levels of the *EGFR* activating mutation, converting to T790M– resistance (f).

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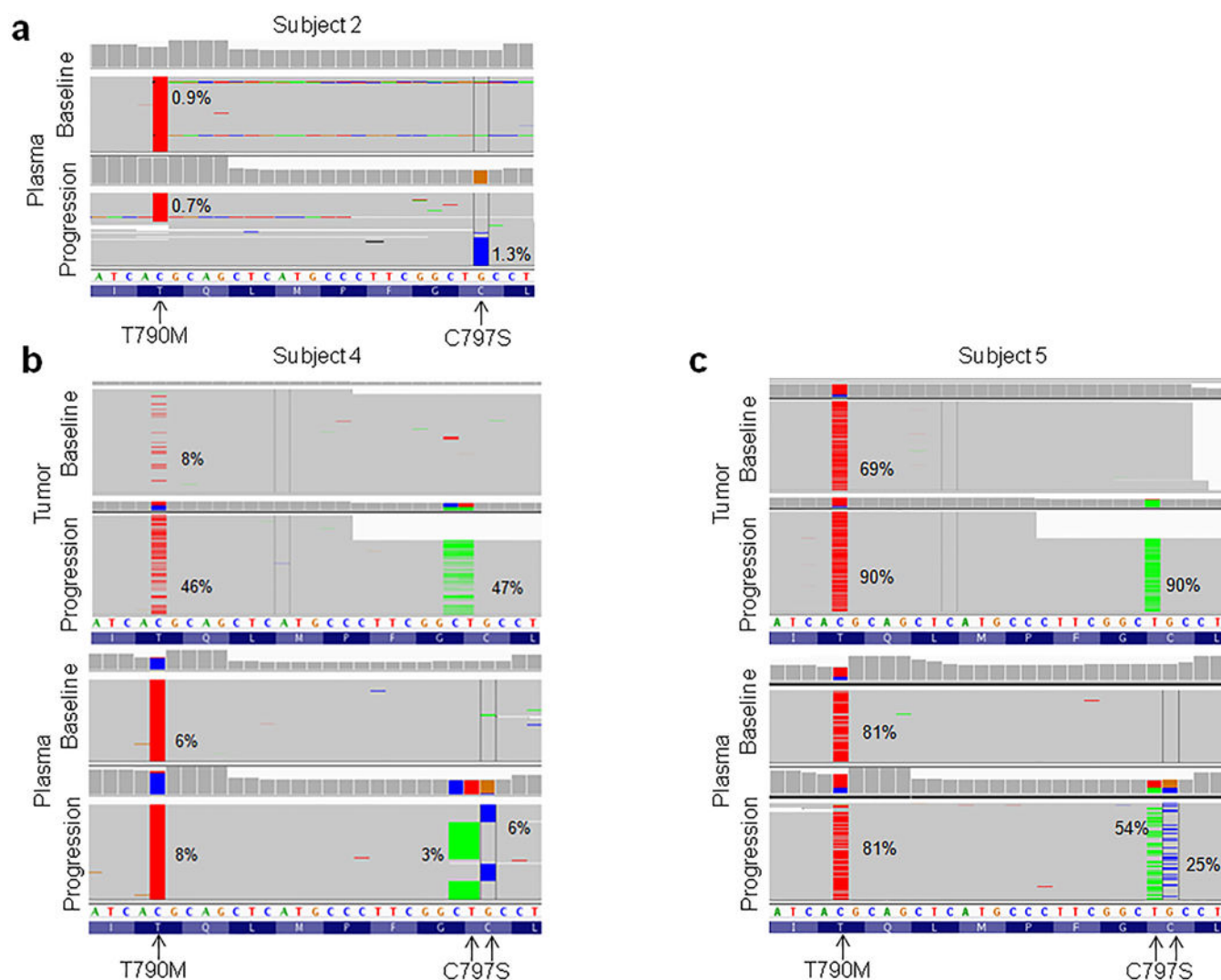


Fig. 2. NGS of baseline and progression cfDNA demonstrates several different genomic presentations of acquired *EGFR* C797S. **(a)** NGS of cfDNA from Subject #2 indicates the acquired C797S G→C mutation (blue) is on a different allele, in *trans* with the T790M mutation. **(b–c)** NGS of baseline and progression tumor biopsies (top panels) confirmed the acquired C797S mutation detected with plasma NGS (bottom). Plasma NGS detects the same T→A C797S mutation (green) found in the tumor and additionally detects a second G→C mutation encoding for C797S (blue).